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PUG TECH TALK · OCTOBER 24, 2026

Agent, Pipeline, or Crew?

How I Choose Multi-Agent Patterns with Microsoft Agent Framework

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HELLO — I'M

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- 15+ years volunteering with Pune User Group
- I write about AI × product strategy, for builders and leaders

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TODAY

From one agent to several, on a laptop

01
Why the pattern matters
Cost, debugging and model size

02
Agent, pipeline, crew
Three patterns, three questions

03
The toolkit
Agent Framework and Foundry Local

04
Live demo
One task, three patterns

05
Q&A
Push back on the taxonomy

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Most “multi-agent systems” are pipelines with a fancier name.

That isn’t pedantry. The label decides what you debug and what you pay.

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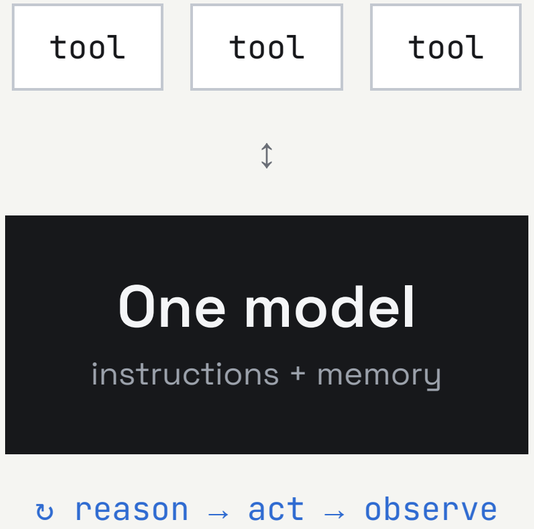
PATTERN 1 OF 3

Agent: one model looping with itself

What it is. One model with instructions and tools. It reasons, calls a tool, reads the result and decides what to do next, until it's done.

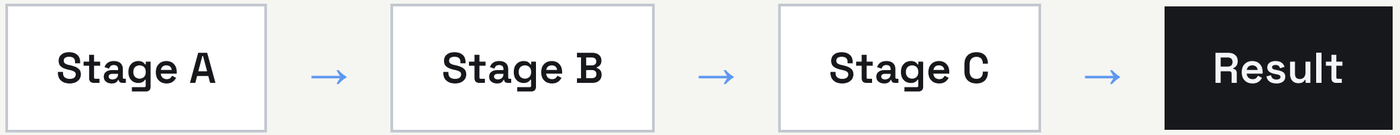
Reach for it when the steps aren't knowable in advance, and one point of view is enough.

The catch: fast and cheap, but opaque. You get an answer, not a trail of why.



PATTERN 2 OF 3

Pipeline: the work only moves forward



What it is. Separately prompted agents, each owning a step and handing its output on. Nobody revisits an earlier stage's work as a peer.

Reach for it when you can name the steps in advance. Most real "multi-agent" products land here.

The payoff: each stage is testable on its own, and a failure points to one step.

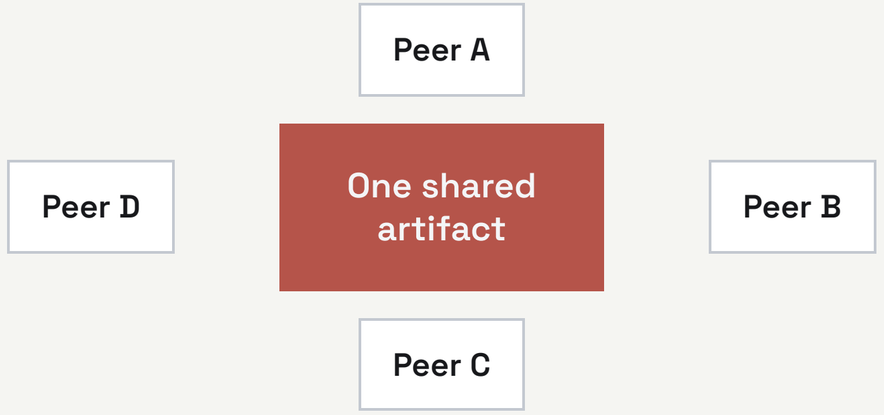
PATTERN 3 OF 3

Crew: peers contest the same artifact

What it is. Agents with different viewpoints argue over one shared output. Order doesn't matter; disagreement is the point.

Reach for it when the disagreement is the product: trade-offs surfaced, not averaged away.

The catch: the most calls, the slowest runs, and models strong enough to rebut, not just restate.



Prior art: Anthropic separates *workflows* from *agents*; CrewAI popularised "crews". These definitions are how I draw the lines.

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WHY IT MATTERS

The pattern decides your bill, your debugging and your model size

01 · COST

Every agent multiplies model calls

Each extra agent, round or retry is another call, and more latency.

02 · DEBUGGING

Failures hide in different places

A pipeline fails at a stage. A crew fails in a conversation.

03 · MODEL SIZE

Not every step needs a big model

Some steps don’t need a model at all. The pattern tells you which.

THE FRAMEWORK

Three questions classify any design

Q1	Does one model loop with itself, or are there separately prompted agents?	→ **One model:** agent Several: next question
Q2	Does the artifact move forward, or get contested at the same level?	→ **Forward:** pipeline **Contested:** crew
Q3	Remove one agent. Does a step disappear, or does the result change character?	→ **A step goes:** pipeline **Character changes:** crew

THE TOOLKIT

Microsoft Agent Framework

Open source, for agents and multi-agent workflows, in Python and .NET

Agents

A model, instructions and tools behind one interface

Workflows

Graphs of agents and plain functions, with checkpoints and human-in-the-loop

Orchestrations

Five built-in multi-agent patterns, ready to use

Any model

Foundry, Azure OpenAI, OpenAI, Ollama, Foundry Local: swap the chat client

DevUI

A local UI to run and debug agents and workflows

OpenTelemetry

Traces for models, agents and workflows, built in

THE TOOLKIT

Foundry Local: the models run on this laptop

- Runs models on your own device: no Azure account, no per-token cost, works offline
- An OpenAI-compatible endpoint, so Agent Framework talks to it like any model
- Picks the right build for your hardware: GPU, NPU or CPU
- A curated catalog of small models such as Phi and Qwen; each reports whether it supports tool calling

```
# install (macOS)
brew tap microsoft/foundrylocal
brew install foundrylocal

# what can this machine run?
foundry model list

# is the service up?
foundry status
```

DEMO BACKLOG · TANDEM

Six features to rank

F1

Home-screen widget

A phone home-screen widget showing today's shared notes and a quick-capture button.

F2

SSO and audit log

SAML single sign-on plus an admin audit trail of who viewed and edited each note.

F3

Offline conflict fix

Stops notes edited offline on two devices from silently overwriting each other.

F4

Smart paste cleanup

Strips stray formatting when pasting from email or web pages into a note.

F5

AI thread summary

One-click summary of a long comment thread on a note, with open questions highlighted.

F6

Admin usage dashboard

A dashboard for workspace admins showing active users and storage per team.

THE DEMO TASK

One task, three ways: rank a backlog

Six features for **Tandem**, a fictional team note-taking app, built so the lenses genuinely disagree.

RICE

Reach × impact × confidence ÷ effort

Strategic fit

Does it move the company's themes?

Effort & risk

What does it cost, and what could go wrong?

Customer pain

How much does it hurt the people asking?

Agent: one model, all four lenses

Pipeline: four scorers, then merge

Crew: four lenses argue it out

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LIVE DEMO

Agent → Pipeline → Crew

Watch the output, the number of model calls, and what happens when one agent leaves.

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IF YOU REMEMBER ONE SLIDE

01 Start with one agent.

02 Move to a pipeline when you can name the steps.

03 Use a crew only when the disagreement is the product.

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Questions?

Push back on the taxonomy. That's how it gets better.

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THANK YOU

Let’s connect.

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